

ABISHEK MURUGAN

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PROFESSIONAL SUMMARY

Physics graduate with specialized Data Science training from the IITM Pravartak-incubated GUVI program. Proficient in Python, SQL, machine learning, and deep learning, with hands-on experience building end-to-end solutions in predictive modeling, computer vision, and behavioral analytics. Experienced in ML pipelines, feature engineering, model evaluation, MLOps, and cloud deployment using Azure. Seeking an entry-level Data Scientist or Machine Learning Engineer role.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Analysis & Visualization: Pandas, NumPy, Matplotlib, Seaborn, Plotly, Exploratory Data Analysis

Machine Learning: Scikit-Learn, XGBoost, Regression, Classification, Clustering, Feature Engineering, Model Evaluation

Deep Learning: PyTorch, Neural Networks, CNN, RNN, LSTM, Transformers, Hugging Face

Cloud & Deployment: Microsoft Azure, ADLS Gen2, Azure Blob Storage, Azure Machine Learning, Streamlit

MLOps & Tools: MLflow, Git, GitHub, Linux

PROJECTS

Intelligent Browser Behavior Analytics Platform

[[GitHub](#)]

Python | Pandas | SQLite | psutil | PyTorch | LSTM | K-Means | Azure ML | MLflow | Streamlit

- Built an end-to-end analytics platform that collects browser activity from the system SQLite database and captures system RAM utilization using psutil for behavioral analysis.
- Engineered event sessionization by grouping browsing activity using a **15-minute inactivity threshold**, enabling session-level behavioral analysis and downstream machine learning.
- Performed correlation analysis between browsing activity and RAM utilization and applied **clustering to identify distinct behavioral patterns** across browsing sessions.
- Developed an **LSTM-based sequence model in PyTorch** using the previous five sessions to predict the next session, with **MLflow experiment tracking and Azure ML-based pipeline execution** supporting the production pipeline and Streamlit analytics dashboard.

Diamond Dynamics — Diamond Price Prediction & Market Segmentation

[[GitHub](#)]

Python | RAPIDS cuML | XGBoost | PyTorch | K-Means | Streamlit | YAML | Astral-uv | Git

- Built an end-to-end machine learning portal for **diamond price prediction and market segmentation** using XGBoost regression and K-Means clustering.
- Trained models using **NVIDIA RAPIDS cuML and XGBoost** on GPU, with the XGBoost model achieving an **R² of 0.9819**.
- Developed a custom GPU –CPU model serialization pipeline to load RAPIDS-trained artifacts for CPU inference without GPU libraries or retraining.
- Developed a **Streamlit inference application** with centralized YAML configuration, modular prediction components, logging, and custom exception handling.

Aero Eye AI — Aerial Object Classification & Detection

[[GitHub](#)]

Python | PyTorch | CNN | ResNet50 | YOLOv8 | Computer Vision | Streamlit | Git

- Built an end-to-end computer vision system for **binary classification and object detection of birds and drones** using a dataset of 3,319 aerial images.
- Developed a lightweight **Custom CNN and a ResNet50 transfer learning model** for bird-versus-drone classification, with ResNet50 achieving approximately **93% test accuracy**.
- Fine-tuned **YOLOv8 Nano** for aerial object detection and localization, achieving **98.3% mAP@50** on the validation set.
- Developed a Streamlit application for image inference, providing classification predictions, confidence scores, bounding-box visualization, model latency metrics, and training diagnostics.

WORK EXPERIENCE

Front Office Manager | Full-time

2024 - 2026

Slam Lifestyle and Fitness Studio | On-site • Coimbatore, India

- Analyzed membership and revenue data using **Excel**, developing sales charts and visualizations to track performance and identify trends.
- Collaborated with the sales team to analyze historical sales trends and optimize sales strategies, contributing to an increase in monthly revenue from ₹7 lakh to ₹10 lakh (**43%**).
- Managed customer onboarding, membership renewals, and day-to-day operations to support customer retention and business performance.

Technologies / Skills Used: Excel, Microsoft Office, Google Workspace

EDUCATION

Sri Ramakrishna College of Arts & Science Coimbatore, India

2022 - 2025

B.Sc. Physics

CERTIFICATIONS

Master Data Science Program – GUVI

Mar 2026

Certified Professional in Advanced Programming - IITM Pravartak

Jul 2026